MATHS

Q 1) → Monomial.

Q 2) → Let $p(x) = (x+1)(x^2 - x - x^4 + 1)$

$$= x^3 - x^2 - x^5 + x^2 - x - x^4 + 1$$

$$= x^3 - x^5 - x^4 + 1$$

The heighest pocoer of $x = 5$.

Page No.:
Date:
YOUVA

Q5)

$$2x^2 - 3x + k = 0$$

$$a = 2,\ b = -3,\ c = k$$

So, $\alpha \times \frac{1}{\alpha} = \frac{c}{a}$

$$1 = \frac{k}{2}$$

$$\boxed{k = 2}$$

Q6)

Sum of its zeroes = 5

product of its zeroes = −14.

$\alpha + \beta = 5$,

$\alpha \times \beta = -14$

$(x-\alpha)(x-\beta)$

$x^2 - \alpha x - \beta x + \alpha\beta$

$x^2 - (\alpha+\beta)x + \alpha\beta$

$x^2 - 5x + (-14)$ $(\because \alpha\beta = -14)$

$= x^2 - 5x - 14$

Q7) →

$$x^2 + 5x + 6 = 0$$

$$(a = 1,\ b = 5,\ c = 6)$$

$$x^2 + 2x + 3x + 6 = 0$$

$$x(x+2) + 3(x+2) = 0$$

$$(x+2)(x+3) = 0$$

$$x + 2 \qquad x + 3$$

$$x = -2 \qquad x = -3$$

$$\alpha = -2 \quad , \beta = -3$$

$$\alpha + \beta = \frac{-b}{a}$$

$$-2 - 3 = \frac{-5}{1}$$

$$-2 - 3 = -5$$

$$-5 = -5$$

$$\alpha\beta = \frac{c}{a}$$

$(-2)(-3) = \frac{6}{1}$

$6 = 6$.

Q8) $\rightarrow$ The zeroes are:- 5, 3, -2.

$(x+3)(x+5)(x-2)$

Simplifying,

$(x-3)(x-5)(x+2)$

$= (x^2-5x+3x+15)(x+2)$

$= (x^2-8x+15)(x+2)$

$= (x^3-8x^2+15x+2x^2-16x+30)$

$= x^3-6x^2-x+30$.

Q 9)

$x+\frac{1}{x}=3$; $x^3+\frac{1}{x^3}=?$

Cubing on both side,

$$\left[x+\frac{1}{x}\right]^3=(3)^3$$

$$x^3+\frac{1}{x^3}+3\cancel{x}\times\frac{1}{\cancel{x}}\left[x+\frac{1}{x}\right]=27$$

$(\because (a+b)^3=a^3+b^2+3ab(a+b)$

$$x^3+\frac{1}{x^3}+3(3)=7$$

$$x^3+\frac{1}{x^3}=27-9$$

$$\boxed{x^3+\frac{1}{x^3}=18}$$